



Materials Science 2: Processing and Applications Revision

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Revision Part 1



What have we learned in this module

- **Metal** materials and processing
- **Polymer** materials and processing
- **Ceramic** materials and processing

Classification of Metals

Metals are divided into 2 families or classes

Ferrous metals

- Carbon steel
- Alloy steel
- Cast iron



What are the differences between steels and cast irons?
What do we use them for?

Non-Ferrous metals

- Aluminium alloys
- Copper alloys
- Titanium alloys
- Nickel alloys
- Refractory metals, noble metals,



What properties do these alloys have that we cannot get from steels?
What do we use them for?

Strengthening of Metals



There are 4 ways of changing the strength of metals:

- **Cold work**
- **Grain size changes**
- **Alloying and phase transformation**
- **Precipitation hardening**

How do we control these to get the properties we want?

What processing methods and heat treatments can we use to achieve the properties we want?

Metal processing

Casting

- Sand cast
- Die cast
- Investment cast
- Welding
- Additive manufacture



Melting and solidification

What effect do these have on the microstructure?
What products are these used for?

Forming operations

- Forging
- Rolling
- Extrusion



Plastic deformation

What shapes can we make with these processes?

Heat treatment

- Annealing
- Tempering



How do these change the microstructure?

Classification of Polymers

There are many ways to classify polymers:

- Thermoplastic
- Thermoset
- Elastomer
- Chemistry (side groups)
- M_w and structure / crystallinity
- Thermal behaviour (T_g , T_m)
- Mechanical behaviour

Which will crystallize?

Which will show a T_g ?

What effect do the side groups and the chain stiffness have on the properties?

Mechanical Properties of Polymers

There are many ways to classify polymers:

- Brittle
- Plastic
- Elastomeric

- Time dependence
- Temperature dependence

Why do polymers have such a wide range of behaviour?

How do these change with T_g ?

What are the mechanisms of deformation in the polymers?

How can we use the processing to control the microstructure and properties?

Polymers Processing

Polymers to products:

- Additives
- Compounding
- Casting
- Thermo-forming
- Films, foams and fibres
- Additive manufacturing

Why do we need to put additives in polymer products?

How will the processing change the microstructure and properties?

Which product shapes can be made from the different processing methods?

Which processing methods are suitable for thermoplastics, thermoset or elastomers?

Ceramic Materials

In-organic, non-metallic materials

- Engineering ceramics
- Functional Ceramics
- Abrasives and refractory
- Glass
- Porcelain
- Cement
- Carbon

What are the differences between these materials?

What do we use them for?

Why are they important?

- Ionic – covalent bonding (mixture)
- Ionic size, stoichiometry and charge balance
- Ceramic crystal structures
- Amorphous materials

What effect do these have on the properties of the materials?

Processing Ceramic Materials

Glass forming processes

- Glass blowing
- Float glass
- Residual stress



Melting or softening

How do we make glass easy to form?

Why are residual stresses a problem?

Particulate forming processes

- Hydroplastic forming
- Ball milling
- Sintering



Non-melting processes

Why do we need to sinter ceramics?

Why is it important to remove porosity and impurities?

Mechanical Properties of Ceramics

Elastic properties

- Elastic modulus

Fracture strength

- Distribution of strength
- Test geometry

Cracking behaviour

- Origin of failure
- Influence of flaws

What are the differences between ceramics and metals or polymers?

What are the differences between crystalline ceramics and glasses?

Processing and Applications

Crystal structures

Phase behaviour

Heat treatments

Processing methods

How are these different for

- Metals
- Polymers
- Ceramics

When are they similar?

What are the differences?

Time to review and revise

What questions do you have?

Revision Part 2



What have we learned in this module

- **Electrical properties** of materials
- **Thermal properties** of materials

Electrical properties

Electrical conduction

- Ohm's law
 - Resistance
 - Resistivity
 - Conductivity
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- Electronic structure of solids
 - Electron mobility
 - Conductivity of -
 - metals
 - ceramics
 - polymers

How are these properties defined?

What is the origin of the behaviour?

What are the ranges of values for the different families of materials?

How do these properties change with temperature, impurities and defects?

Semi-conductor materials and their electrical behaviour:

- Band gap structure
- Electron – hole pairs
- Intrinsic semi-conductors
- Extrinsic semi-conductors
- Charge carrier mobility

What do we mean by these terms?

What is the origin of the behaviour?

What are the ranges of values for the different families of materials?

- Diodes
- Junction transistors
- MOS-FETs

How do we make useful devices from semi-conductor materials?

Dielectric materials

Dielectric materials and their electrical behaviour:

- Capacitance
- Permittivity
- Dielectric constant
- Dielectric strength

How are these properties defined?

What are the ranges of values for the different families of materials?

- Types of polarization
- Dielectric displacement
- Ferro-electric behaviour
- Piezo-electric behaviour

What is the origin of the behaviour?

Applications of electric properties

How are electrical materials used in engineering products:

Piezo-electric materials

Capacitors

Non-volatile memory

CMOS technology

How are the conductor, semi-conductor and dielectric properties of materials used to make devices?

How can we select the correct materials for different applications?

Thermal properties

Important thermal properties of materials:

- Heat capacity
and specific heat capacity
- Thermal conductivity
mechanisms of conduction
- Thermal diffusivity

How are these properties defined?

What is the origin of the behaviour?

What are the ranges of values for the different families of materials?

What are the mechanisms of heat conduction for the different families of materials?

Thermo-mechanical behaviour

Important thermal properties of materials:

- Thermal expansion (CTE)
- Thermal stresses
- Differential expansion
- Thermal shock
- Thermal management

How are these properties defined?

What is the origin of the behaviour?

What are the ranges of values for the different families of materials?

How can we select materials for different applications where thermal properties are important?

Revision Part 3



What have we learned in this module

- **Magnetic properties** of materials
- **Optical properties** of materials

Learning outcomes

Magnetic properties:

- Origin of magnetic behaviour
- Classification of different magnetic responses
- Characterisation of magnetic hysteresis
- Applications of magnetic materials

Optical properties important technologies:

- Interaction of light waves with the electronic structure of solids
- Optical response of materials that results in:
transparency, opacity and colour
- Technologies that use optical behaviour

Magnetic properties

Definition of magnetic parameters:

- Magnetic dipoles
- Applied magnetic field, H
- Induced magnetic field, B
- Permeability
- Magnetic susceptibility
- Magnetization

How are these properties defined?

What is the origin of the behaviour?

What is the relationship between these parameters?

Magnetic properties

Types of magnetism:

- Diamagnetic
- Paramagnetic
- Ferromagnetic
- Ferrimagnetic
- Antiferromagnetic

What do these terms mean?

What are the magnetic responses with and without an applied magnetic field?

What magnitude is the magnetic susceptibility for each of these?

Which materials have these magnetic responses?

Which are useful for magnetic material applications?

Magnetic properties

Types of magnetic behaviour:

- Dipole domains
- Hysteresis loops
- Hard magnetic behaviour
- Soft magnetic behaviour
- Applications of hard and soft magnetic material
- Superconduction

What do these terms mean?

How can we characterise the behaviour?

How do we use the behaviour in magnetic materials applications?

Optical properties

Types of optical behaviour:

- Electro-magnetic radiation
- Energy of photons
- Interaction with the electronic structure of solids for different materials families

How does EM radiation interact with the electronic structure of solids?

What energy exchanges are possible?

How are these different for the different families of materials?

How do they result in the optical behaviour that we see?

Optical properties

Types of optical behaviour:

- Reflected
- Absorbed
- Luminescent
- Transmitted
- Refracted
- Scattered

What do these terms mean?

What is the origin of the behaviour?

What optical response can we expect from different materials?

Optical properties

Technologies that use optical behaviour:

- LEDs
- Photo-voltaics
- Lasers
- Optical fibres

Which optical behaviours do they use?

How do we make useful devices?

What advantages do they provide?

Revision Part 4



What have we learned in this module

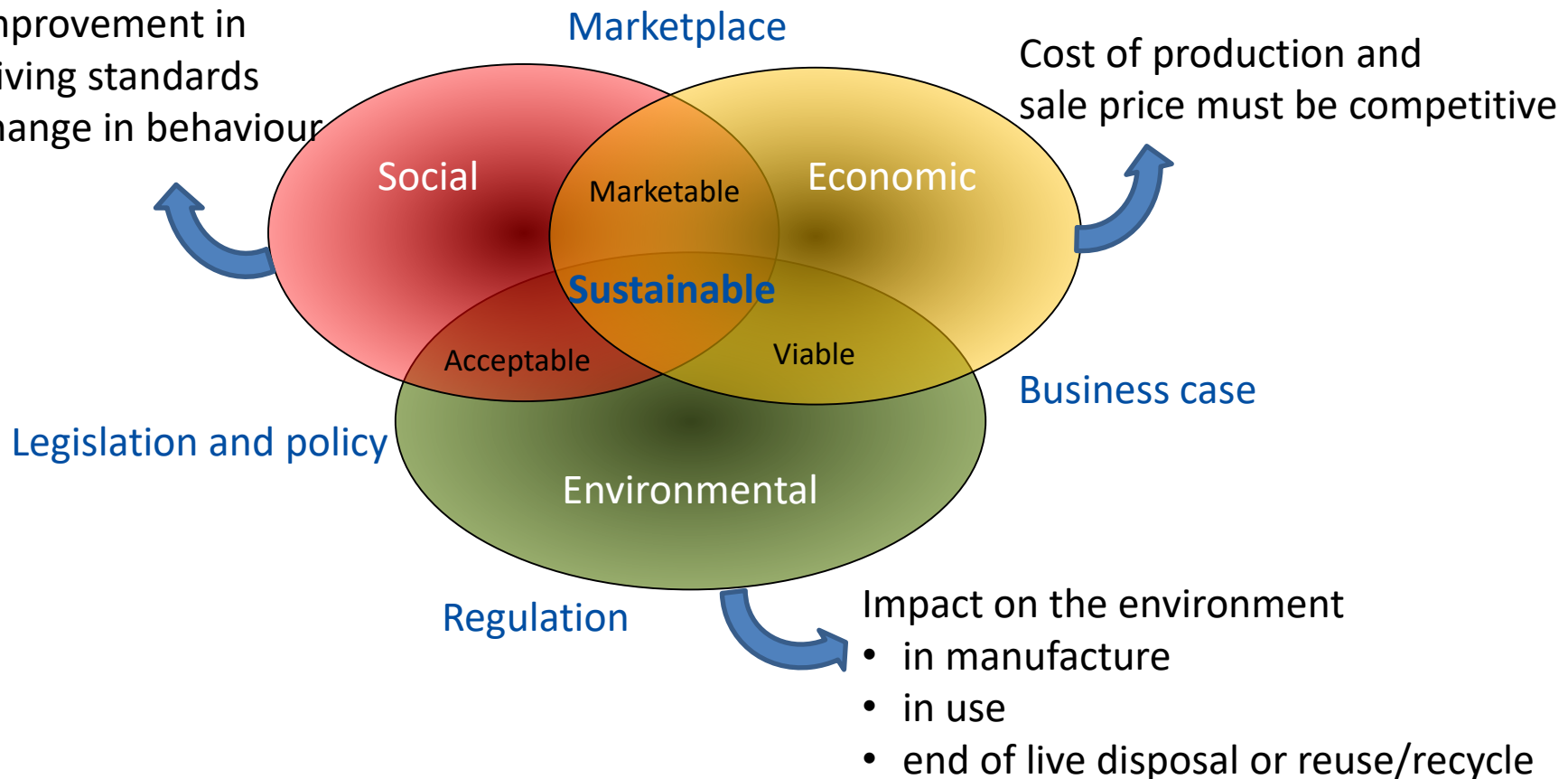
- **Sustainable engineering**

Sustainable engineering

Sustainable engineering requires consideration of 3 factors :

Impact on society

- improvement in living standards
- change in behaviour



Sustainable engineering

What do we mean by 'sustainable engineering'?

- Environmental, societal and economic impact of engineering products at all stages of their lifetime from 'Cradle to Grave'

Life Cycle Analysis

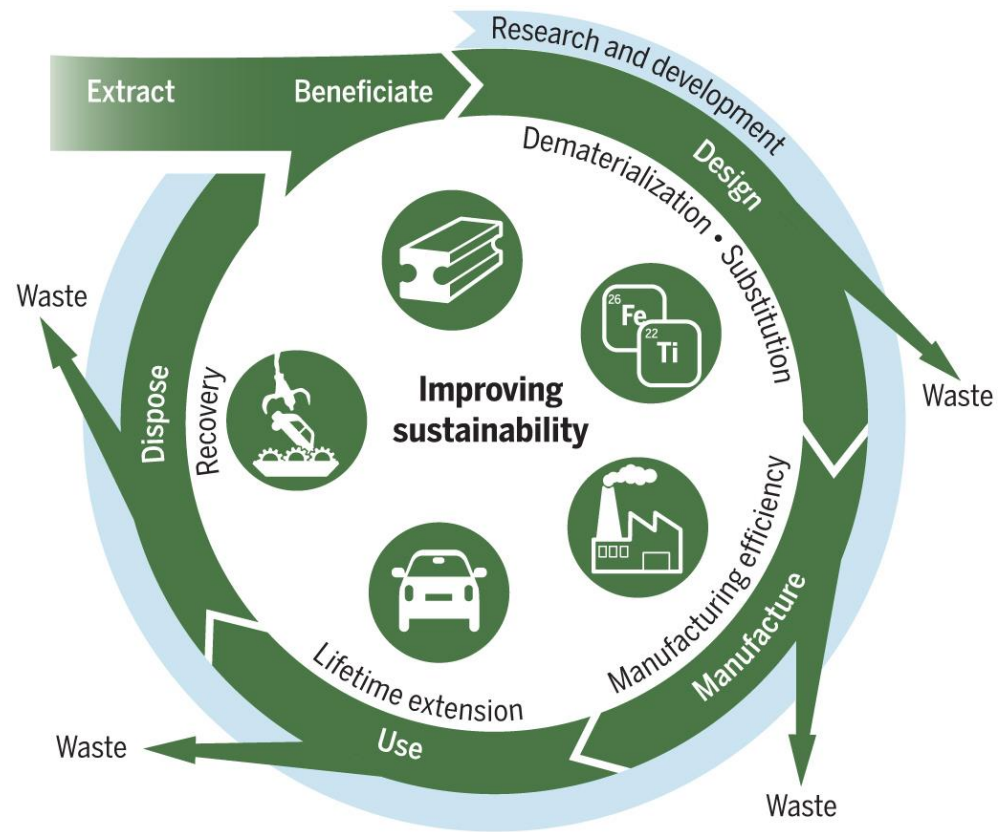
- Inputs and outputs at different stages of the product life cycle
- Resources
- Energy
- Generation of waste

Definitions of terms

- Embodied energy
 - Carbon footprint
 - Process energy
 - End-of-life potential
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- Reduce
 - Reuse
 - Recycle
-
- E-waste
 - Right to repair

Design for Environmental Sustainability

- Design for environment means consider environmental impact from concept stages
- Selection of raw material based on sustainable resources
- Consider properties based on microstructure
- Include processing to improve sustainability
- Close the loop including upcycle and recycle



Environmental impact

Assessment of impact at each stage

- Materials
- Production
- Distribution
- Use
- Recovery

Identify where impact can be reduced

- Reduce energy
- Reduce waste

Consider examples in materials processing

Consider examples in product design and manufacture